

# Text-Aligned Radar-Based Sign Language Recognition for Healthcare Communication

Raffaele Mineo<sup>1\*</sup>

raffaele.mineo@unict.it

Amelia Sorrenti<sup>1\*</sup>

amelia.sorrenti@phd.unict.it

Gaia Caligiore<sup>2</sup>

gaia.caligiore@unimore.it

Federica Proietto Salanitri<sup>1</sup>

federica.proiettosalanitri@unict.it

Giovanni Bellitto<sup>1</sup>

giovanni.bellitto@unict.it

Senya Polikovsky<sup>3</sup>

senya@tue.mpg.de

Sabina Fontana<sup>1</sup>

sabina.fontana@unict.it

Egidio Ragonese<sup>1</sup>

egidio.ragonese@unict.it

Concetto Spampinato<sup>1</sup>

concetto.spampinato@unict.it

Simone Palazzo<sup>1</sup>

simone.palazzo@unict.it

<sup>1</sup> University of Catania, Italy

<sup>2</sup> University of Modena and Reggio Emilia, Italy

<sup>3</sup> Max Planck Institute for Intelligent Systems, Germany

## Abstract

Sign language recognition (SLR) aims to bridge the communication gap between signers and non-signers through automated interpretation of signs. While most existing methods rely on RGB or depth video, we explore radar signals as a privacy-preserving alternative. We introduce TRACE (Text-Radar Alignment via Contrastive Embedding), a novel radar-based framework for SLR. TRACE extracts spatiotemporal embeddings from Range-Doppler Map (RDM) sequences using a 3D convolutional neural network (CNN). During training, it employs prompt-conditioned contrastive learning to align radar features with soft-prompted class embeddings from a frozen text encoder. At inference time, TRACE assigns to each radar sequence the label whose textual embedding is most similar to the visual representation. We evaluate our method on a radar-based Italian Sign Language (LIS) dataset, showing that it outperforms state-of-the-art baselines under standard classification metrics. The results and the ablation study confirm that joint training enhances embedding quality by guiding the CNN toward semantically meaningful features through text alignment, while learnable soft prompts yield more suitable textual representations.

## 1. Introduction

Sign language recognition (SLR) is a research area focused on the automatic interpretation of signs to support communication between Deaf or hard-of-hearing individuals and non-signers. In recent years, this field has seen significant progress, particularly in the area of *isolated sign recognition*, where each sign is processed independently. Early approaches relied on handcrafted features and shallow classifiers, with limited capacity to generalize across signers or vocabularies [26, 35]. The advent of deep learning allowed for end-to-end modeling of the spatiotemporal dynamics inherent in signing, leveraging RGB and depth video to capture both manual and non-manual components [6, 20].

Despite their success, camera-based systems introduce concerns in privacy-sensitive environments such as hospitals or rehabilitation centers. In these contexts, continuous video monitoring may raise ethical and legal issues, and performance may degrade under suboptimal lighting or cluttered backgrounds [18, 22]. Radar-based sensing has recently emerged as a promising privacy-preserving alternative. Frequency-modulated continuous wave (FMCW) radar systems, in particular, offer robust motion capture without revealing identifiable visual information [13, 19], and are resilient to lighting conditions and occlusions.

Previous radar-based sign recognition approaches have demonstrated the feasibility of capturing dynamic movements using micro-Doppler signatures and other radar-

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\*Equal contribution.

derived features [9, 15, 29]. However, these works often focus on American Sign Language (ASL), targeting small and constrained vocabularies [2, 27]. This narrow focus limits their applicability to other sign languages, which differ significantly in lexicon, grammar, and articulation. Moreover, existing radar-based systems generally lack mechanisms for integrating semantic priors from natural language, missing the opportunity to guide the learning process through textual supervision.

In this work, we introduce **TRACE** (*Text-Radar Alignment via Contrastive Embedding*), a novel radar-based framework for isolated SLR. Our method leverages Range-Doppler Map (RDM) sequences from FMCW radar to extract spatiotemporal representations through a 3D convolutional neural network. To incorporate semantic grounding, we employ a frozen text encoder augmented with learnable soft prompts [36], enabling prompt-conditioned embeddings of class labels. During training, TRACE optimizes a contrastive objective that aligns visual radar features with corresponding textual embeddings, encouraging the model to learn semantically meaningful representations.

We evaluate our method on the MultiMeDaLIS dataset [24], which contains radar recordings of isolated LIS signs specifically curated for clinical and assistive communication contexts. Experimental results demonstrate that TRACE outperforms state-of-the-art radar-based baselines on standard classification metrics. Ablation studies confirm the importance of multimodal contrastive alignment and prompt-based conditioning in improving generalization and feature discriminability.

The contributions of this paper are as follows:

- We propose TRACE, a novel privacy-aware architecture for radar-based isolated SLR that aligns radar sequences with textual class descriptions via contrastive learning.
- We introduce prompt-conditioned textual embeddings using a frozen language model, enabling adaptable and semantically grounded classification.
- We evaluate our method on a radar-acquired LIS dataset, demonstrating improved performance and robustness over prior methods.

Our findings highlight the potential of combining radar data with language supervision for sign recognition in privacy-critical environments, supporting the development of assistive technologies and clinical applications.

## 2. Related work

Sign language recognition (SLR) has garnered significant attention in recent years, particularly in the context of *isolated sign recognition*, where individual signs are classified from separate video segments. Early approaches primarily relied on handcrafted features and traditional classifiers, often limited in their ability to generalize across signers or vocabularies.

Deep learning has enabled effective modeling of the spatiotemporal patterns in sign gestures by allowing real-time recognition of both manual and non-manual features from either RGB or depth data. Pigou et al. [26] employed convolutional neural networks to recognize isolated signs using RGB and depth data, demonstrating high accuracy on small vocabularies. Wang et al. [35] proposed a compact representation of temporal dynamics using depth maps, achieving strong performance in the ChaLearn LAP challenge [12]. Li et al. [20] explored the use of temporal graph convolutional networks applied to 2D body pose sequences, offering an alternative to vision-based architectures. Boháček et al. [6] further advanced pose-based recognition by introducing SPOTER, a Transformer-based model that leverages 2D keypoints from the face, hands, and body.

Recent work has also focused on modeling sign language articulation through multimodal datasets and architectures that combine visual, skeletal, and motion-based cues. The AUTSL dataset [30], focused on Turkish Sign Language, introduced a benchmark with RGB, depth, and skeleton modalities. Maruyama et al. [23] proposed a three-stream network that integrates global motion, local hand/face appearance, and skeletal dynamics, achieving state-of-the-art results on the MS-ASL [17] and WLASL [20] benchmarks. Similarly, De Coster et al. [10] and Vahdani et al. [33] introduced systems that fuse body pose flow with hand-cropped RGB frames, or combine multiple 3D CNNs across modalities, respectively.

However, in clinical environments, the use of cameras introduces privacy concerns and can suffer from performance drops due to background clutter or poor lighting [18, 22]. To overcome these limitations, radar-based approaches have emerged as a promising alternative, demonstrating robustness to lighting variations and background clutter [13, 19]. Radar-based systems, such as those using micro-Doppler or frequency-modulated continuous wave (FMCW) signals, can capture fine-grained motion without revealing identifiable visual content [13, 15]. Chen et al. [9] enhanced classification accuracy by fusing features from multiple domains (e.g., range, velocity, and angle), eliminating the need for visual sensors. Ren et al. [29] addressed continuous sign recognition by segmenting FMCW radar signals using amplitude thresholding and classifying signs based on Doppler-derived features.

Other approaches have explored radar and sensor fusion for sign and motion recognition, without relying on visual data. Rahman et al. [27, 28] explored two approaches to mitigate kinematic discrepancies between imitation and native signing in RF-based ASL recognition: adversarial domain adaptation with 77 GHz FMCW radar and kinematically-constrained generative models for data synthesis. Jhaung et al. [16] introduced a dynamic hand sign recognition system using a BiLSTM-based model ap-

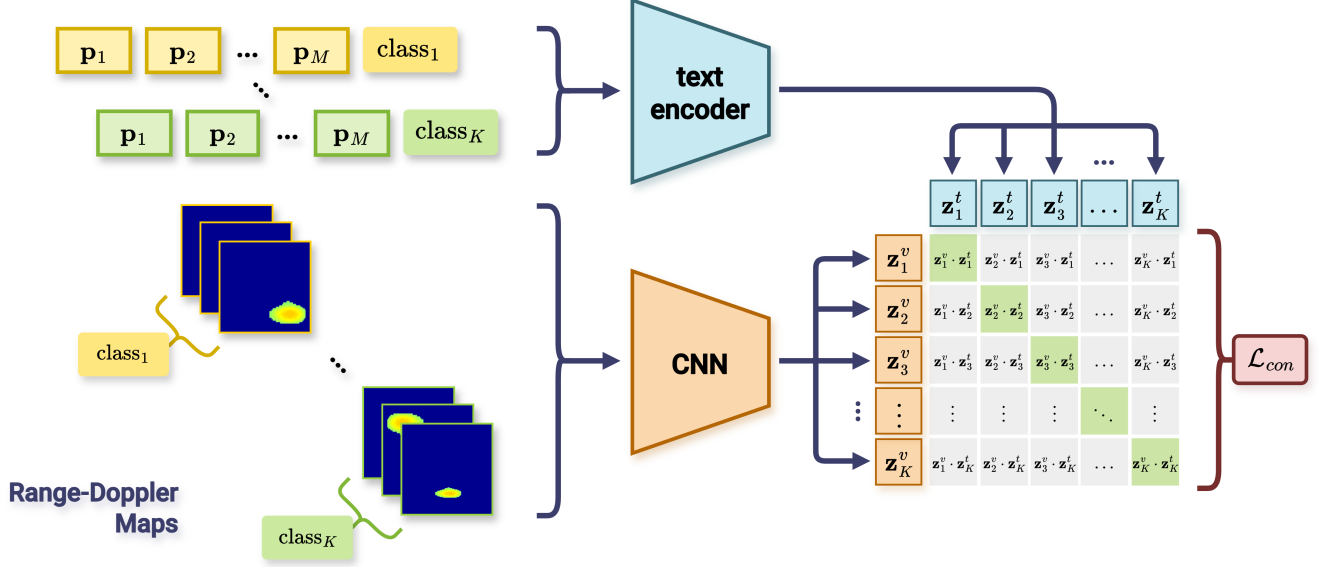


Figure 1. **Overview of the TRACE method.** Radar-based Range-Doppler Map (RDM) sequences are encoded into visual embeddings  $\mathbf{z}_i^v$  using a 3D CNN. Each class label is represented textually and prepended with learnable prompts before being processed by a frozen text encoder, yielding text embeddings  $\mathbf{z}_k^t$ . A contrastive loss  $\mathcal{L}_{\text{con}}$  encourages alignment between matching visual-text pairs in the shared embedding space.

plied to range-Doppler maps from 60 GHz radar. Debnath et al. [11] fused FMCW radar range profiles and micro-Doppler signatures to classify traffic gestures. Arab et al. [3] proposed a dual-branch CNN that processes 24 GHz radar signals and continuous wavelet transform (CWT) spectrograms for human motion recognition. Beyond these, HRSpecNET [4] reconstructs high-resolution micro-Doppler signatures from 1-D radar signals, while PLFNets [5] employ structured complex-valued filterbanks for direct classification from raw IQ data.

Despite their robustness, the application of radar-based methods remains limited: most studies focus on generic signs or small American Sign Language (ASL) vocabularies, relying on modest datasets with constrained lexicons [2, 22]. Broader and more diverse datasets are needed to support deep learning models in recognizing complex sign repertoires across various sign languages and to address domain-specific communicative needs, such as doctor-patient interaction in clinical contexts. While several Italian Sign Language (LIS) corpora have been developed for linguistic or educational purposes [1, 34], many are either private, limited in size, or centered on continuous signing. Consequently, they are poorly suited for training deep learning models for isolated SLR, particularly in medical contexts where isolated commands are prevalent [7, 21].

Our approach builds on these insights by using radar signals from the MultiMeDaLIS dataset [24], processed into range-Doppler maps, to develop a privacy-aware recognition system for Italian Sign Language.

### 3. Method

To better capture domain-specific patterns in Italian Sign Language, the proposed method leverages the CLIP text encoder with learnable prompts and employs a 3D convolutional network applied to radar-derived range-Doppler maps as its visual encoder. Figure 1 illustrates an overview of the proposed method.

#### 3.1. Problem formulation

Let  $\mathcal{D} = \{(\mathbf{x}_i, \mathbf{y}_i)\}$  be a dataset, where  $\mathbf{x}_i \in \mathcal{X}$  is an input sample and  $\mathbf{y}_i \in \mathcal{Y}$  is its corresponding class label. Each input sample  $\mathbf{x}_i$  is a sequence of frames  $\{\mathbf{f}_{i,1}, \mathbf{f}_{i,2}, \dots, \mathbf{f}_{i,T_i}\}$ , where  $T_i$  is the temporal length of the video. The goal of our work is to develop a model that maps any given input sample  $\mathbf{x}_i \in \mathcal{X}$  to its corresponding class label  $\mathbf{y}_i \in \mathcal{Y}$ .

#### 3.2. Text encoding

Each class label  $\mathbf{y}_i \in \mathcal{Y}$  is a short textual string, corresponding to either a medical term or a letter of the Italian alphabet in LIS.

Let  $\mathbb{V}$  be a fixed vocabulary; each string is tokenized into  $\mathbb{V}^N$ , a sequence of words from  $\mathbb{V}$ . To simplify notation, we treat  $N$  as variable-length, leveraging the length-agnostic nature of transformer-based text encoders enabled by padding tokens. The resulting token sequences  $\mathbb{V}^N$  are then projected into a continuous vector space through a text embedding function  $\varepsilon_t : \mathbb{V}^N \rightarrow \mathbb{R}^{N \times d}$ , where  $d$  denotes the embedding size. For simplicity, we assume that each class

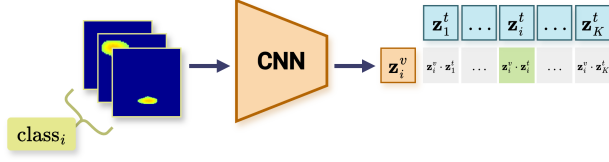


Figure 2. **Overview of the inference phase.** The CNN encodes the RDM sequence into a visual embedding  $\mathbf{z}_i^v$ , which is compared with all text embeddings  $\mathbf{z}_k^t$  corresponding to class labels. The predicted class is the one with the highest cosine similarity between  $\mathbf{z}_i^v$  and  $\mathbf{z}_k^t$ .

label  $\mathbf{y}_i$  has already been tokenized,  $\mathbf{y}_i \in \mathbb{V}^N$ .

A pretrained text encoder  $\phi_t : \mathbb{R}^{(N+M) \times d} \rightarrow \mathbb{R}^d$  is employed to project a tokenized sequence to a numeric representation of size  $d$ . We then define a set  $P$  of  $M$  learnable soft prompts,  $P = \{\mathbf{p}_1, \dots, \mathbf{p}_M\}$ .

When extracting the text representation for class  $k$ , rather than simply computing  $\phi_t(\varepsilon_t(\mathbf{y}_k))$ , we prepend the soft prompts to the embedded class label before passing it to the encoder, resulting in:

$$\mathbf{z}_k^t = \phi_t([\mathbf{p}_1, \dots, \mathbf{p}_M, \varepsilon_t(\mathbf{y}_k)]). \quad (1)$$

Since the class labels may vary across different semantic categories (e.g., body parts, medical actions, and pathological conditions), the use of prompt-based conditioning is motivated by the observation that the text encoder might not fully capture the intended semantics of domain-specific terms. By conditioning on learnable prompts, we enable the encoder to adapt its representations in a parameter-efficient manner, allowing it to extract more discriminative features of the target concept.

### 3.3. RDM encoding and classification

Given an input sequence  $\mathbf{x}_i$  consisting of  $T$  frames of size  $H \times W$ , a network  $\phi_v : \mathbb{R}^{T \times H \times W} \rightarrow \mathbb{R}^d$  encodes it to a  $d$ -dimensional latent representation as follows:

$$\mathbf{z}_i^v = \phi_v(\mathbf{x}_i). \quad (2)$$

This network serves as the visual encoder, replacing the conventional image-based encoder used in CLIP with a 3D convolutional network tailored to process radar-derived temporal sequences.

During training, the visual representation  $\mathbf{z}_i^v$  is aligned with its corresponding textual representation  $\mathbf{z}_i^t$  through a contrastive loss  $\mathcal{L}_{\text{con}}$ , which encourages high cosine similarity between matching visual-text pairs and penalizes mismatches within the batch. This objective allows the model to project the RDM sequences into a shared embedding space consistent with their semantic labels.

The overall loss function is defined as:

$$\mathcal{L}(\theta_v, P) = \mathbb{E}_{(\mathbf{x}_i, \mathbf{y}_i) \sim \mathcal{D}} [\mathcal{L}_{\text{con}}(\mathbf{z}_i^v, \mathbf{z}_i^t)] \quad (3)$$

where  $\mathbf{z}_i^t$  and  $\mathbf{z}_i^v$  are obtained using eq. 1 and eq. 2, respectively.

As shown in Fig. 2, classification during inference is performed by computing the cosine similarity between the visual embedding  $\mathbf{z}_i^v$  and all textual class embeddings  $\mathbf{z}_k^t$ , selecting the class with the highest similarity score.

## 4. Experimental evaluation

In this section, we present the experimental evaluation of our method. Section 4.1 and Section 4.2 describe the dataset and the preprocessing pipeline. We then outline the evaluation metrics (Sec. 4.3) and the training procedure (Sec. 4.4). Results and the ablation study are presented in Section 4.5 and Section 4.6, respectively.

### 4.1. Dataset

The MultiMeDaLIS dataset [24] is a digital collection of Italian Sign Language (LIS) with a specific focus on the medical domain. It comprises 126 distinct items, including 100 health-related signs and the 26 letters of the Italian alphabet, designed to support applications in patient-doctor communication contexts. The signs are semantically categorized into body parts, symptoms and conditions, medical procedures, and healthcare-related concepts. They are further organized into semantically marked and unmarked macro-categories, reflecting their relevance and specificity within the medical context.

For this study, we rely exclusively on radar data due to its ability to preserve privacy, capture fine hand motion, and remain unaffected by lighting conditions or the need for physical contact. The 60 GHz millimeter-wave sensor used (Infineon XENSIV BGT60TR13C) samples at 13 frames per second and captures the three-dimensional movement and positioning of the signer's hands and fingers. It produces time-domain log-scale data from three separate antennas, enabling detailed spatiotemporal analysis of sign articulation.

The dataset includes a total of 25,830 radar sign instances, corresponding to 205 repetitions for each of the 126 signs. Each sign was performed in isolation by a single signer with substantial experience in LIS, starting and ending in a standardized neutral posture.

The resulting dataset offers a vision-free representation of LIS signs in the medical domain, ensuring privacy protection by relying exclusively on radar signals.

### 4.2. Data preprocessing

The raw radar signals acquired from the 60 GHz mmWave sensor undergo a structured preprocessing pipeline to compute the Range-Doppler Map (RDM) representations. Figure 3 presents a simplified overview of this preprocessing pipeline, from raw radar data to final RDM representation.



Figure 3. **Overview of the radar signal preprocessing pipeline leading to the Range-Doppler Map (RDM).** The diagram shows the main stages of the processing chain, including range FFT, Doppler FFT, and thresholding. While simplified, the figure highlights the key transformation from raw time-domain radar data to a 2D RDM representation. Intermediate operations such as mean removal, windowing, zero-padding, and FFT shifting are omitted for clarity.

Each frame consists of time-domain logarithmic-scale data recorded via one transmit antenna and three receive antennas (RX), sampled at 1 MHz with a transmit power level of 31 and an IF gain of 40 dB. Under these parameters, the radar achieves a range resolution of 0.0312 m and a speed resolution of 0.0321 m/s, with a maximum detectable range of approximately 1.60 m and a maximum speed of 4.11 m/s. Each frame is acquired at 0.077 s intervals, centered at 60.50 GHz, allowing for fine-grained capture of hand gestures typical of sign language.

To derive the range-Doppler spectrum, we apply a two-stage FFT process on the time-domain radar data. First, a range FFT is performed over the fast-time (i.e., chirp) dimension after optional mean removal, Blackman-Harris windowing, and zero-padding. The resulting matrix is then transposed, and a Doppler FFT is applied over the slow-time (i.e., chirps) dimension using a Chebyshev window, followed by further zero-padding. The complex 2D output is converted to an absolute magnitude spectrum, and an FFT shift is applied to center the DC component. A configurable spectral thresholding step suppresses low-magnitude values (default of -90 dB, tunable between -75 dB and -120 dB), enhancing signal clarity by reducing background noise.

An optional Moving Target Indication (MTI) filter can be applied to further suppress static background signals by zeroing out range-Doppler bins below a predefined velocity threshold. Both the standard and MTI-filtered range-Doppler maps have a resolution of  $128 \times 1024$ . Figure 4 illustrates the visual difference between an RDM with and without MTI filtering.

All signal processing steps are designed to preserve the spatiotemporal fidelity of the signing gestures, ensuring that the resulting RDM representations retain the fine-grained motion patterns necessary for accurate recognition.

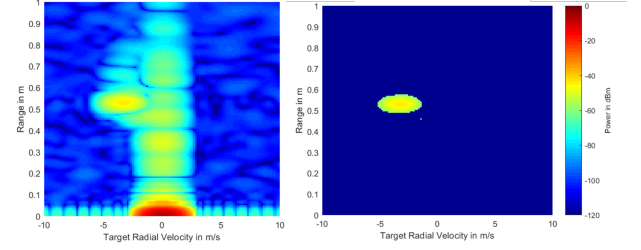


Figure 4. **Range-Doppler map (RDM) with and without MTI filtering.** Comparison between the same RDM without MTI filtering (left) and with MTI filtering applied (right). The MTI filter suppresses static background signals by zeroing out bins below a predefined velocity threshold, resulting in a cleaner representation of dynamic hand motion.

### 4.3. Metrics

To assess the performance of our sign recognition model, we employ standard classification metrics, including *accuracy*, *sensitivity* (recall), *specificity*, and the  $F_1$  score. These metrics are computed from the confusion matrix, which summarizes the counts of true positives, true negatives, false positives, and false negatives. For brevity, the  $F_1$  score is denoted as *F-score* in the tables.

### 4.4. Training procedure

We adopt a 5-fold nested cross-validation, with outer folds for testing and inner folds for validation. The visual encoder is a 3D convolutional backbone, optionally initialized from Kinetics-400. Visual features and text embeddings are projected into a common embedding space, L2-normalized, and compared via scaled cosine similarity, with a learnable logit scale initialized to  $\log(1/0.07)$ .

We train for 30 epochs with a batch size of 12, minimizing class-balanced cross-entropy with per-class weights inversely proportional to class frequency. Optimization uses AdamW with a learning rate of 0.0001 and weight decay of 0.001. Gradient clipping and automatic mixed precision are enabled, and a plateau scheduler optionally adjusts the learning rate based on validation performance.

### 4.5. Results

To evaluate the effectiveness of our proposed method, we compare it against a range of state-of-the-art baselines, including RGB, RGB-D, and radar-based sign language recognition approaches. For methods without publicly available code, we implemented the models ourselves based on the original descriptions, reproducing them to the best of our ability. All models were evaluated using their default hyperparameters, with model selection guided by validation performance. The results are reported in Table 1, using the evaluation metrics described in Sec. 4.3.

| Method                | Data type                                                | Acc.                             | Sens.                            | Spec.                            | F-score                          |
|-----------------------|----------------------------------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|
| De Coster et al. [10] | RGB                                                      | 88.4 $\pm$ 3.7                   | 78.6 $\pm$ 3.4                   | 98.2 $\pm$ 3.5                   | 79.7 $\pm$ 3.9                   |
| Vahdani et al. [33]   | RGB-D                                                    | 84.1 $\pm$ 2.9                   | 83.1 $\pm$ 4.1                   | 85.1 $\pm$ 2.1                   | 84.2 $\pm$ 2.2                   |
| Caligiore et al. [8]  | RGB-D                                                    | 74.6 $\pm$ 3.9                   | 66.4 $\pm$ 3.1                   | 82.8 $\pm$ 3.0                   | 59.9 $\pm$ 3.3                   |
| Jhaung et al. [16]    | Radar                                                    | 71.9 $\pm$ 3.3                   | 61.9 $\pm$ 2.3                   | 81.9 $\pm$ 3.4                   | 69.2 $\pm$ 2.9                   |
| Debnath et al. [11]   | Radar                                                    | 79.3 $\pm$ 3.5                   | 83.3 $\pm$ 3.3                   | 75.3 $\pm$ 3.1                   | 79.3 $\pm$ 3.9                   |
| Arab et al. [3]       | Radar                                                    | 81.0 $\pm$ 3.4                   | 65.9 $\pm$ 3.1                   | 96.1 $\pm$ 2.8                   | 29.6 $\pm$ 2.9                   |
| Mineo et al. [25]     | 3 $\times$ RDM + 3 $\times$ MTI                          | 93.6 $\pm$ 0.5                   | 87.9 $\pm$ 0.5                   | 99.3 $\pm$ 0.6                   | 87.7 $\pm$ 0.5                   |
| <b>TRACE (ours)</b>   | <b>3<math>\times</math>RDM + 3<math>\times</math>MTI</b> | <b>99.3 <math>\pm</math> 0.1</b> | <b>98.8 <math>\pm</math> 0.2</b> | <b>99.9 <math>\pm</math> 0.1</b> | <b>98.8 <math>\pm</math> 0.2</b> |

Table 1. **Comparison with state-of-the-art methods in sign language recognition.** The “Data type” column indicates the primary sensor modality employed by the method. The best performance is achieved by TRACE, which combines 3 $\times$ RDM and 3 $\times$ MTI inputs, highlighting the importance of multimodal radar encoding and multi-perspective information. TRACE leverages the MixConv architecture pretrained on radar data and employs 64 soft prompts, yielding the best overall performance among all methods.

Among existing approaches, RGB- and RGB-D-based models often achieve strong performance in isolated sign recognition tasks. However, these methods rely on visual input, which may raise privacy concerns in sensitive environments such as hospitals or rehabilitation centers. Moreover, their performance can degrade under challenging conditions such as variable lighting or cluttered backgrounds, limiting their applicability in real-world scenarios.

Despite being privacy-preserving, prior radar-based methods [3, 11, 16] have generally underperformed compared to their vision-based counterparts. This performance gap can be attributed to the use of handcrafted features or shallow architectures, which fail to capture the rich spatiotemporal structure of sign articulation. Jhaung et al. [16] and Arab et al. [3] report  $F_1$  scores below 70%, while Debnath et al. [11] achieves 79.3% by incorporating additional radar features.

Our method, which combines three antenna inputs with both Range-Doppler Map (RDM) and Moving Target Indicator (MTI) representations, overcomes these limitations and achieves superior performance across all metrics. The integration of RDM and MTI provides richer motion cues that, together with spatiotemporal modeling and prompt-conditioned contrastive learning, enhance the model’s ability to capture sign dynamics. This is further supported by a comparison with Mineo et al. [25], who use the same radar dataset. While their method achieves strong overall results, TRACE achieves higher performance, demonstrating the benefits of multimodal radar encoding and language-aligned supervision.

#### 4.6. Ablation study

To assess the impact of each component in our model, we perform a series of experiments focusing on three key as-

| Data type                                                | Acc.                             | Sens.                            | Spec.                            | F-score                          |
|----------------------------------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|
| RDM                                                      | 95.2 $\pm$ 0.4                   | 94.6 $\pm$ 0.2                   | 95.9 $\pm$ 0.7                   | 93.8 $\pm$ 0.2                   |
| 3 $\times$ RDM                                           | 97.9 $\pm$ 0.3                   | 97.5 $\pm$ 0.4                   | 98.3 $\pm$ 0.2                   | 97.9 $\pm$ 0.1                   |
| MTI                                                      | 92.0 $\pm$ 0.3                   | 90.4 $\pm$ 0.8                   | 93.6 $\pm$ 0.3                   | 90.4 $\pm$ 0.5                   |
| 3 $\times$ MTI                                           | 92.4 $\pm$ 0.2                   | 90.6 $\pm$ 0.4                   | 94.3 $\pm$ 0.1                   | 91.3 $\pm$ 0.8                   |
| RDM + MTI                                                | 97.8 $\pm$ 0.4                   | 97.4 $\pm$ 0.6                   | 98.2 $\pm$ 0.2                   | 97.3 $\pm$ 0.3                   |
| <b>3<math>\times</math>RDM + 3<math>\times</math>MTI</b> | <b>99.3 <math>\pm</math> 0.1</b> | <b>98.8 <math>\pm</math> 0.2</b> | <b>99.9 <math>\pm</math> 0.1</b> | <b>98.8 <math>\pm</math> 0.2</b> |

Table 2. **Impact of radar input types and number of antennas on model performance.** We evaluate the effect of using RDM, MTI, and their combination, in both single- and multi-antenna configurations. The best performance is achieved by combining 3 $\times$ RDM and 3 $\times$ MTI inputs, highlighting the importance of multimodal radar encoding and multi-perspective information.

pects: radar input configuration, backbone architecture, and the number of soft prompts used in textual supervision.

**Radar input types and antenna configuration.** Table 2 evaluates the effect of radar modality and antenna configuration, comparing RDM, MTI, and their combination under both single- and multi-antenna setups. Results show that combining RDM and MTI outperforms single-modality inputs, as the two provide complementary information: RDM encodes spatial and velocity cues, while MTI focuses on motion saliency. In addition, multi-antenna input offers diverse perspectives that enhance robustness. The best results are obtained using 3 $\times$ RDM and 3 $\times$ MTI, confirming the effectiveness of both modality fusion and multi-view radar encoding.

| Architecture              | Acc.                             | Sens.                            | Spec.                            | F-score                          |
|---------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|
| ResNet 3D                 | 93.5 $\pm$ 0.7                   | 87.9 $\pm$ 0.4                   | 99.0 $\pm$ 0.9                   | 90.3 $\pm$ 0.4                   |
| $\hookrightarrow$ Pre-tr. | 98.0 $\pm$ 0.1                   | 96.1 $\pm$ 0.2                   | 99.9 $\pm$ 0.1                   | 96.8 $\pm$ 0.6                   |
| R(2+1)D                   | 88.1 $\pm$ 2.8                   | 85.3 $\pm$ 3.4                   | 90.9 $\pm$ 2.2                   | 87.4 $\pm$ 2.6                   |
| $\hookrightarrow$ Pre-tr. | 97.5 $\pm$ 0.4                   | 95.0 $\pm$ 0.8                   | 99.9 $\pm$ 0.1                   | 95.9 $\pm$ 0.3                   |
| MixConv                   | 93.2 $\pm$ 0.6                   | 86.8 $\pm$ 0.9                   | 99.6 $\pm$ 0.3                   | 85.7 $\pm$ 2.0                   |
| $\hookrightarrow$ Pre-tr. | <b>99.3 <math>\pm</math> 0.1</b> | <b>98.8 <math>\pm</math> 0.2</b> | <b>99.9 <math>\pm</math> 0.1</b> | <b>98.8 <math>\pm</math> 0.2</b> |

Table 3. **Effect of backbone architecture and pretraining on performance.** We compare ResNet 3D [14], R(2+1)D [32], and MixConv [31] with and without pretraining on radar data. Results show that both architectural choice and pretraining substantially impact performance, with the pretrained MixConv achieving the highest scores across all metrics.

| # prompts | Acc.                             | Sens.                            | Spec.                            | F-score                          |
|-----------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|
| 0         | 90.3 $\pm$ 0.2                   | 86.3 $\pm$ 0.7                   | 94.4 $\pm$ 0.3                   | 89.0 $\pm$ 0.3                   |
| 4         | 92.8 $\pm$ 1.1                   | 89.1 $\pm$ 0.7                   | 96.4 $\pm$ 2.8                   | 92.9 $\pm$ 0.3                   |
| 8         | 93.6 $\pm$ 1.4                   | 93.8 $\pm$ 1.2                   | 93.3 $\pm$ 3.9                   | 93.0 $\pm$ 0.8                   |
| 16        | 96.6 $\pm$ 0.1                   | 94.3 $\pm$ 1.2                   | 98.8 $\pm$ 1.1                   | 95.2 $\pm$ 0.3                   |
| 32        | 97.8 $\pm$ 0.6                   | 97.0 $\pm$ 0.2                   | 98.6 $\pm$ 1.0                   | 98.5 $\pm$ 0.3                   |
| 48        | 98.5 $\pm$ 0.5                   | 98.2 $\pm$ 0.2                   | 98.7 $\pm$ 0.7                   | 98.3 $\pm$ 0.4                   |
| <b>64</b> | <b>99.3 <math>\pm</math> 0.1</b> | <b>98.8 <math>\pm</math> 0.2</b> | <b>99.9 <math>\pm</math> 0.1</b> | <b>98.8 <math>\pm</math> 0.2</b> |

Table 4. **Effect of the number of soft prompts on classification performance.** We evaluate the effect of varying the number of learnable prompts prepended to each class label. Results show that using 64 prompts leads to the best performance across all metrics, highlighting the benefit of richer prompt-based conditioning for textual supervision.

**Backbone architecture and pretraining.** In Tab. 3, we compare three backbone architectures: ResNet 3D [14], R(2+1)D [32], and MixConv [31], each evaluated with and without pretraining on radar data. The results show that pretraining consistently improves performance, highlighting the value of prior exposure to radar-specific spatiotemporal patterns. Among the tested models, pretrained MixConv achieves the best overall results, suggesting that its mixed convolutional structure is particularly effective in capturing the heterogeneous motion dynamics of sign language.

**Prompt-based textual supervision.** Table 4 reports the effect of varying the number of soft prompts used in the text encoder. Performance consistently improves as the number of prompts increases, with only marginal gains at 4 and 8 but substantial improvements from 16 onwards. The best performance across all metrics is achieved with 64 prompts. These results support our hypothesis that, since class labels span heterogeneous semantic categories, the frozen text encoder alone may not fully capture their intended meaning.

Prompt-based conditioning introduces additional learnable context that helps disambiguate and align domain-specific terms with radar features, thereby enhancing classification performance.

## 5. Conclusion

In this work, we introduced TRACE, a novel framework for radar-based sign language recognition that leverages prompt-conditioned contrastive learning to align spatiotemporal radar features with textual class embeddings. Unlike traditional vision-based approaches, our method operates on range-Doppler sequences, enabling privacy-preserving recognition that remains robust to lighting and background conditions.

Through a comprehensive evaluation on the Multi-MeDaLIS dataset, TRACE achieved state-of-the-art results, surpassing both radar- and vision-based baselines. This performance stems from the effective integration of multi-modal radar representations, pretrained spatiotemporal encoders, and prompt-conditioned textual supervision, which together enhance the model’s ability to capture the motion patterns of sign language.

Our findings highlight the potential of radar sensing as a viable and privacy-aware modality for sign language understanding, particularly in sensitive domains such as clinical and assistive communication. Future work will explore the extension of TRACE to continuous sign recognition and cross-lingual generalization across different sign languages.

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